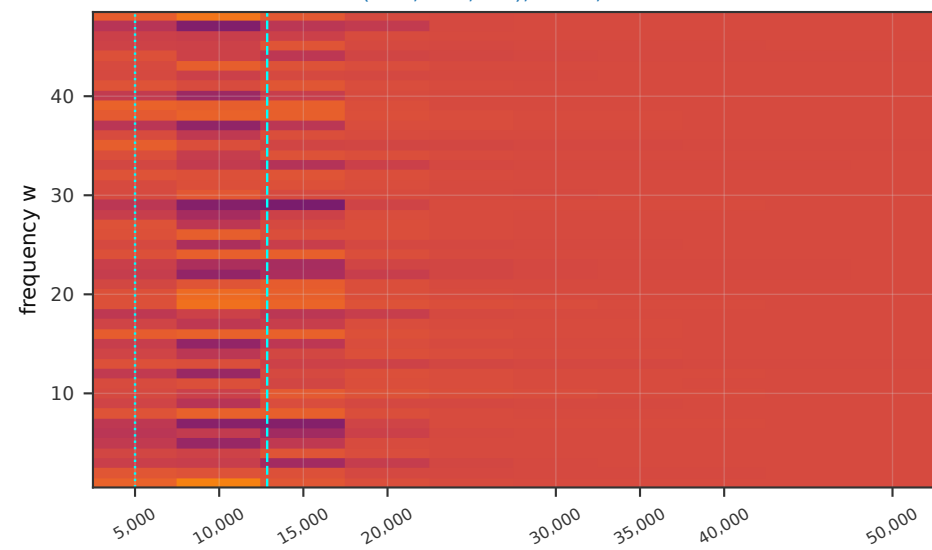
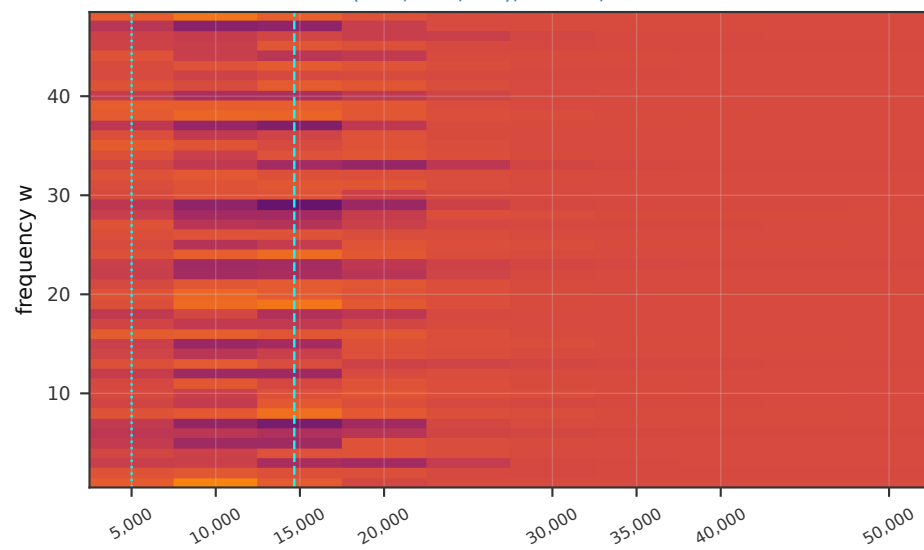


Clock-frequency energy of the logits through training: GD spreads over many frequencies; the transformer concentrates on a few

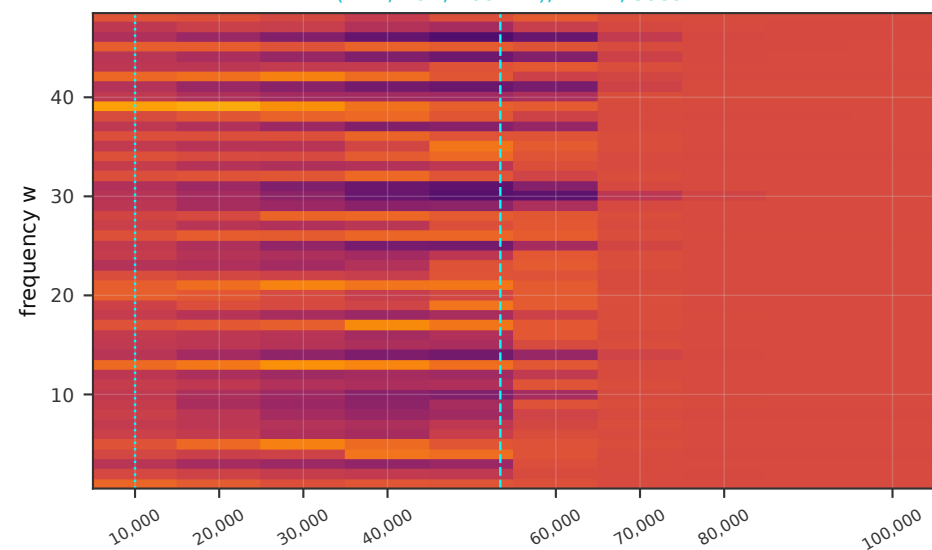
A (MLP, Z97, GD), K = 2, seed 42



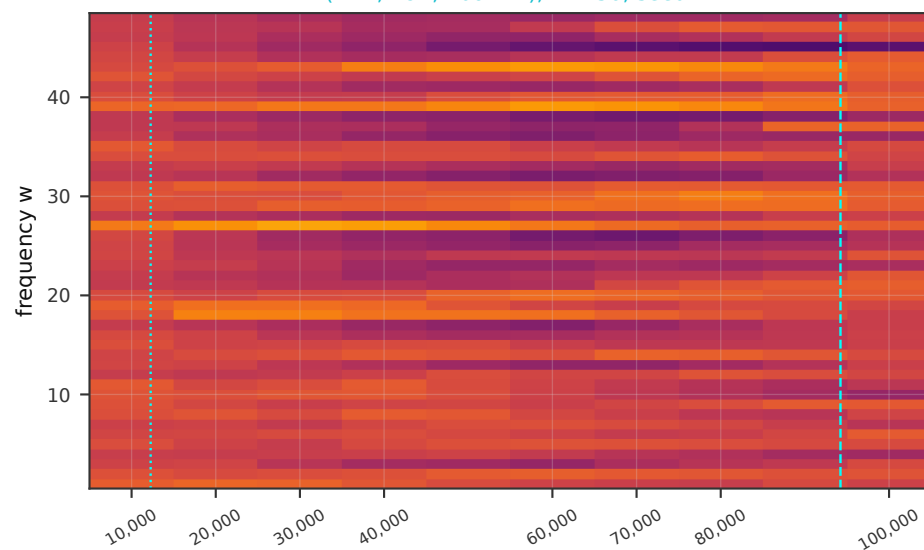
A (MLP, Z97, GD), K = 50, seed 42



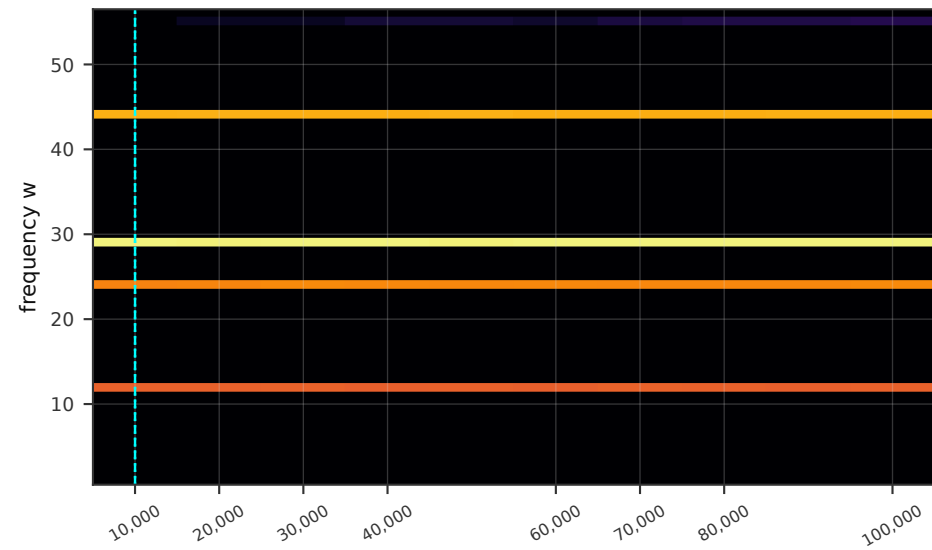
A' (MLP, Z97, AdamW), K = 2, seed 42



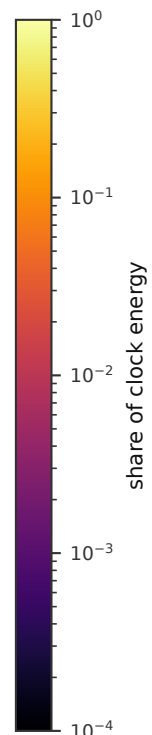
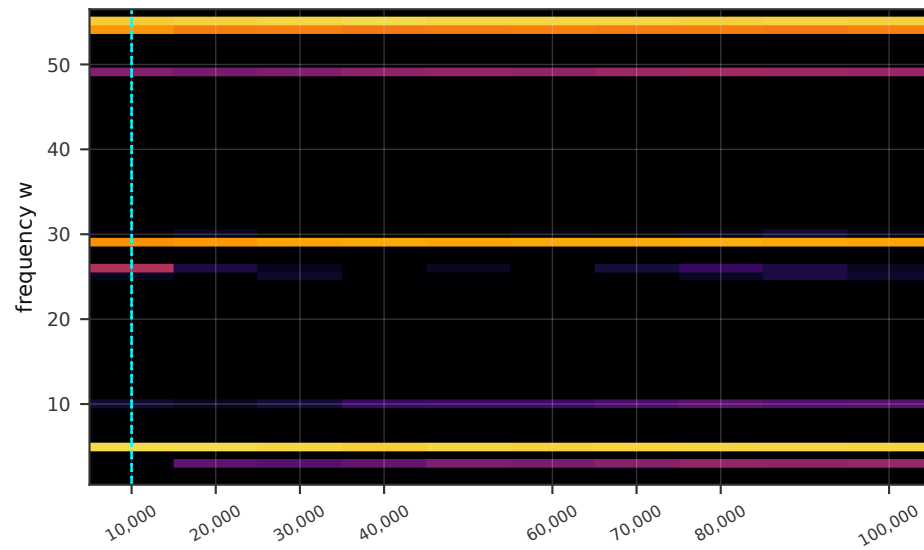
A' (MLP, Z97, AdamW), K = 50, seed 42



B (transformer, Z113), K = 2, seed 42



B (transformer, Z113), K = 20, seed 42



checkpoint step (cyan: t_memo dotted, t_first_cross dashed)

checkpoint step (cyan: t_memo dotted, t_first_cross dashed)